Familial Hyperprolactinemia (PRLR): Comprehensive Disease Characteristics Report

Disease: Familial Hyperprolactinemia · **MONDO:** MONDO:0014250 · **OMIM:** #615555 · **Causal gene: PRLR** (prolactin receptor, HGNC:9446, chromosome 5p13-p12) · **Category:** Genetic (Mendelian endocrine disorder)

Summary

Familial hyperprolactinemia is a rare Mendelian endocrine disorder caused by germline loss-of-function variants in *PRLR*, the gene encoding the prolactin receptor. The prototypical lesion is the autosomal-dominant heterozygous missense variant **p.His188Arg (c.563A>G)**, which disrupts the high-affinity ligand-binding interface of the receptor and abolishes downstream JAK2–STAT5 signaling, producing a state of **prolactin insensitivity (resistance)**. Biallelic inactivating genotypes (e.g., compound-heterozygous Pro269Leu/Arg171*) extend the spectrum to patients with hyperprolactinemia accompanied by agalactia (failure of lactation). Critically, this disorder is tumor-negative: patients have elevated serum prolactin but a normal pituitary MRI**, distinguishing it from the far more common prolactinoma. This report concerns the PRLR (prolactin receptor) form specifically — not the prolactin ligand gene (PRL), nor pituitary-tumor genes (AIP, MEN1).

The mechanistic core is a broken negative-feedback loop. Under normal physiology, prolactin signals through PRLR on hypothalamic tuberoinfundibular dopaminergic (TIDA) neurons, which release dopamine to inhibit pituitary lactotroph prolactin secretion — a short-loop feedback circuit. When PRLR is inactivated, this feedback is lost: dopaminergic inhibitory tone on lactotrophs falls, lactotrophs are disinhibited, and prolactin secretion rises **without any tumor**. Simultaneously, peripheral tissues (breast, gonads, decidua) become resistant to the elevated prolactin, producing the clinical phenotype of oligomenorrhea/amenorrhea, infertility, and agalactia. *PRLR* knockout and heterozygous mouse models strongly corroborate this biology, recapitulating sterility, implantation failure, and gene-dosage-dependent lactation failure.

A striking feature of *PRLR* biology is **bidirectionality**: loss-of-function variants cause familial hyperprolactinemia, whereas **gain-of-function variants** (constitutively active receptors, e.g., p.Ile146Leu, p.Ile76Val) cause the opposite phenotype — benign breast tumors (multiple fibroadenomas) through ligand-independent STAT5 activation and proliferation. This report synthesizes genetics, mechanism, diagnosis, management, epidemiology, and model-organism evidence for the loss-of-function disease, with explicit ontology term suggestions and primary-literature citations throughout. Because the disorder is ultra-rare (only a handful of kindreds described worldwide since 2013), several template sections are necessarily sparse, and those gaps are flagged explicitly.

---

1. Disease Information

Overview. Familial hyperprolactinemia is a hereditary form of non-physiological hyperprolactinemia in which serum prolactin is persistently elevated due to inherited inactivation of the prolactin receptor, *not* due to a prolactin-secreting pituitary adenoma, drug exposure, hypothyroidism, renal failure, or other secondary cause. It is a **diagnosis of exclusion** confirmed by identifying a pathogenic germline *PRLR* variant. Clinically, affected individuals (predominantly women identified to date) present with menstrual irregularity, infertility, and — in biallelic/strongly inactivating genotypes — failure of lactation (agalactia).

Key identifiers:

Resource	Identifier
MONDO	MONDO:0014250
OMIM (phenotype)	#615555 (Hyperprolactinemia)
Gene	*PRLR*, HGNC:9446; NCBI Gene 5618; Ensembl ENSG00000113494; UniProt P16471; OMIM gene *176761
ICD-10	E22.1 (Hyperprolactinemia)
ICD-11	5A60.1 (Hyperprolactinaemia)
MeSH	D006966 (Hyperprolactinemia)
Orphanet	Rare hyperprolactinemia (no dedicated PRLR-specific ORPHA code at time of writing)

Synonyms / alternative names: familial isolated prolactin receptor deficiency; hereditary hyperprolactinemia; familial idiopathic hyperprolactinemia; prolactin insensitivity; prolactin resistance.

Data provenance: Information is derived from **aggregated disease-level resources** (OMIM, PubMed case reports and kindred studies) and functional in-vitro/model-organism work, rather than large EHR cohorts. The disease is defined by a small number of published families; there is no population registry specific to the *PRLR* form.

2. Etiology

Primary cause — genetic. The disease is caused by germline **loss-of-function variants in *PRLR***. Two inheritance architectures are documented:

- **Autosomal dominant** via a heterozygous inactivating missense variant (**p.His188Arg**), which behaves dominantly (haploinsufficiency and/or dominant-negative effect on receptor dimers). This was the founding observation in three affected sisters PMID: 24195502.
- **Autosomal recessive / biallelic** via compound-heterozygous inactivating variants (**Pro269Leu / Arg171***), associated with hyperprolactinemia plus agalactia PMID: 36445946.

Genetic risk factors. The causal variants themselves are the risk factors; no separate susceptibility loci or GWAS signals are established for this Mendelian condition. Modifier genes are unknown.

Environmental risk factors. None established for the *PRLR* form. (For hyperprolactinemia in general, drugs — especially dopamine-antagonist antipsychotics such as paliperidone/risperidone — hypothyroidism, chest-wall stimulation, stress, and renal failure are classic secondary causes, but these are *differentials* to be excluded, not causes of the familial disease.)

Protective factors. None established. No protective *PRLR* alleles are described for this phenotype.

Gene–environment interactions. Not characterized. Physiological states that elevate prolactin (pregnancy, lactation, estrogen exposure) may interact with reduced receptor signaling, but this is inferred rather than demonstrated.

---

3. Phenotypes

Phenotypes reflect both **central disinhibition** (elevated prolactin) and **peripheral prolactin resistance** (end-organ non-response). Because reported kindreds are tiny, frequencies are qualitative.

Phenotype	Type	HPO suggestion	Onset	Severity / frequency	Evidence
Elevated serum prolactin / hyperprolactinemia	Laboratory abnormality	HP:0000870 (Hyperprolactinemia)	Adult (reproductive age)	Defining feature; all affected	PMID: 24195502
Oligomenorrhea / amenorrhea	Clinical sign	HP:0000876 / HP:0000141	Adolescence–adulthood	Common in affected women	PMID: 24195502
Infertility / subfertility	Clinical sign	HP:0000789	Reproductive age	Variable	Human + model PMID: 9009200
Agalactia / failure of lactation	Clinical sign	HP:0100783 (Lactation difficulties)	Postpartum	Seen with biallelic/ strong LOF	PMID: 36445946
Absence of galactorrhea	Distinguishing feature	—	—	Contrasts with prolactinoma	Inferred from resistance

Quality-of-life impact. Chiefly reproductive and psychosocial: infertility, menstrual disturbance, and inability to breastfeed. The condition is **not life-threatening** and does not itself cause mass effect, visual loss, or hypopituitarism (those belong to prolactinoma). No disease-specific EQ-5D/SF-36 data exist for the *PRLR* form.

Note on peripheral resistance paradox: Because end-organs are prolactin-resistant, classic high-prolactin signs that depend on prolactin *action* (e.g., galactorrhea) may be **absent or muted**, even though circulating prolactin is high — a clinically important distinction from prolactinoma.

4. Genetic / Molecular Information

Causal gene. *PRLR* (prolactin receptor), a class-1 cytokine-receptor-superfamily single-pass transmembrane receptor. OMIM gene *176761; HGNC:9446.

Pathogenic variants (documented):

Variant (protein)	cDNA	Zygosity	Classification	Type	Consequence	Phenotype
p.His188Arg	c.563A>G	Heterozygous	Pathogenic	Missense	Loss of function (disrupts ligand binding; abolishes JAK2-STAT5)	Familial hyperprolac (AD)
p.Pro269Leu + p.Arg171*	—	Compound het	Pathogenic	Missense + nonsense	Biallelic inactivation	Hyperprolac + agalactia
p.Ser434Phe	c.1301C>T	Heterozygous	Candidate (missense)	Missense	Proposed hyperprolactinemia association	Familial idio hyperprolac

Gain-of-function counterparts (opposite disease pole — benign breast tumors, NOT hyperprolactinemia): p.Ile146Leu and p.Ile76Val confer constitutive, ligand-independent activity PMID: 18779591, PMID: 19897676. A rare activating p.Asn492Ile has been linked to increased prolactinoma incidence PMID: 36445946.

Allele frequency. The pathogenic loss-of-function variants are extremely rare / private to the described kindreds; they are not common polymorphisms in gnomAD. By contrast, p.Ile76Val and p.Ile146Leu are relatively common in the general population and, in a separate study, had no effect on signaling after prolactin stimulation and were **not** associated with sporadic prolactinoma PMID: 26641246 — underscoring that variant interpretation must be functional, not frequency-only.

Origin. Germline (constitutional); no somatic involvement in this disease.

Functional consequence. Loss of function (His188Arg, Arg171*, Pro269Leu). His188Arg disrupts the high-affinity ligand-binding interface, so prolactin cannot productively engage the receptor, abolishing JAK2–STAT5 phosphorylation.

Modifier genes / epigenetics / chromosomal abnormalities. None established for this disorder. No methylation, histone, or large structural lesions are implicated; this is a single-gene coding-variant disease.

5. Environmental Information

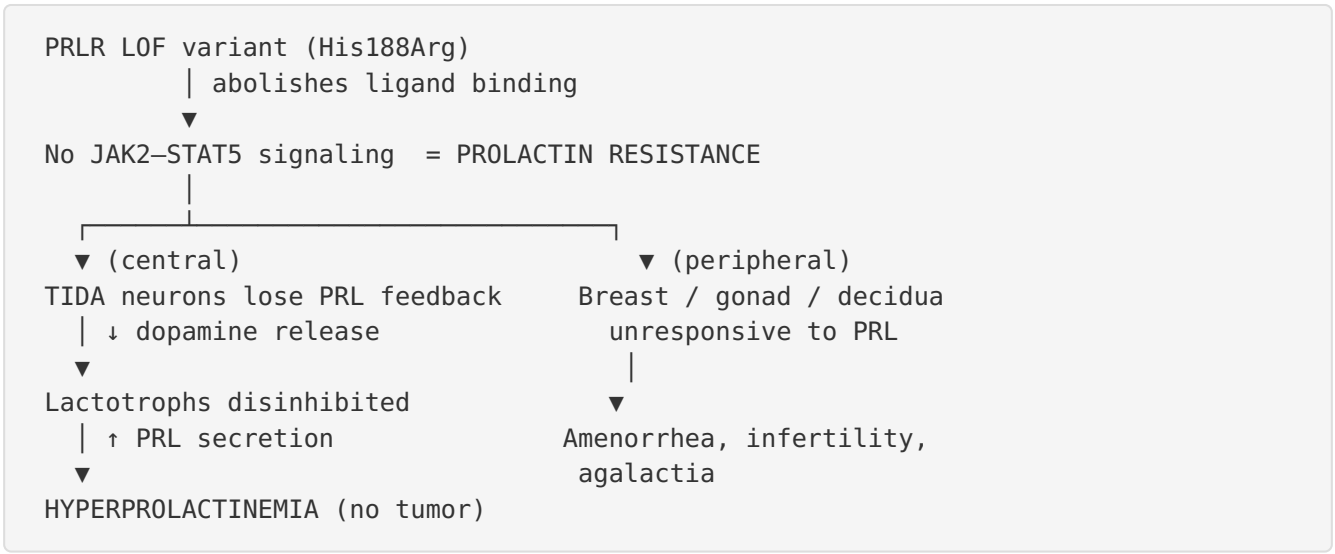
No **environmental, lifestyle, or infectious** agents cause the familial *PRLR* disease — it is monogenic. These categories are relevant only to the **secondary hyperprolactinemias** that must be excluded during workup: dopamine-antagonist drugs (antipsychotics, metoclopramide), estrogens, primary hypothyroidism, chronic kidney disease, chest-wall trauma/stimulation, and stress. (Example drug signal: paliperidone palmitate strongly associated with increased blood prolactin and galactorrhea in pharmacovigilance data, PMID: 39833706.) CHEBI entities relevant to differentials/therapy: dopamine (CHEBI:18243), cabergoline (CHEBI:3286), bromocriptine (CHEBI:3181).

---

6. Mechanism / Pathophysiology

Ordered causal chain

1. **Germline inactivating *PRLR* variant** (e.g., His188Arg) **results in** a prolactin receptor that cannot bind ligand at its high-affinity interface (demonstrated in vitro) [PMID: 24195502].
2. Defective ligand binding **leads to** failure of receptor dimerization/activation and **abolishes** JAK2 autophosphorylation and STAT5 phosphorylation — i.e., **prolactin insensitivity / resistance** (demonstrated in vitro) [PMID: 24195502].
3. **Branch A — Central disinhibition (drives the hyperprolactinemia):** Loss of functional PRLR on hypothalamic **tuberoinfundibular dopaminergic (TIDA) neurons results in** failure of prolactin's short-loop negative feedback. Normally prolactin activates TIDA neurons, which release dopamine to inhibit lactotrophs (demonstrated in mouse) [PMID: 30759390]. Without feedback, **dopaminergic inhibitory tone on pituitary lactotrophs falls** (inferred in humans; demonstrated feedback circuit in mouse).
4. Reduced dopamine **leads to disinhibition of anterior-pituitary lactotrophs**, which **results in increased prolactin secretion → tumor-negative hyperprolactinemia** (inferred for humans, consistent with normal MRI) [PMID: 24195502; 30759390].
5. **Branch B — Peripheral resistance (drives the clinical signs):** Inactive PRLR in **gonadal axis, mammary epithelium, and decidua results in** end-organ non-response to the (now elevated) prolactin → **oligomenorrhea/amenorrhea, infertility, and agalactia** (human phenotype; corroborated by *Prlr*-null/heterozygous mice) [PMID: 9009200; 10935023; 36445946].



Detail by category

- **Molecular pathways:** JAK2–STAT5 is the canonical prolactin pathway (Reactome "Prolactin receptor signaling"; also PI3K-AKT and MAPK contributions). Its loss is the central defect [PMID: 24195502; 10596761].
- **Cellular processes:** Failure of prolactin-driven mammary alveologenesis/differentiation and corpus-luteum support; decidual regulation disrupted (prolactin normally silences IL-6 and 20α-HSD in decidua via JAK2-STAT5) [PMID: 17255200].
- **Protein dysfunction:** Loss of function via disrupted extracellular ligand-binding interface (His188Arg); truncation (Arg171*). No aggregation/misfolding described.
- **Immune/inflammatory, metabolic, oxidative:** Not primary features of the *PRLR* disease.
- **GO term suggestions:** GO:0038161 (prolactin signaling pathway), GO:0007565 (female pregnancy), GO:0007595 (lactation), GO:0046543 (development of secondary female sexual characteristics), GO:0007259 (JAK-STAT cascade).
- **CL term suggestions:** CL:0002313 (lactotroph / prolactin-secreting cell of pituitary), CL:0000700 (dopaminergic neuron / TIDA neuron), CL:0002326 (luminal epithelial cell of mammary gland).

---

7. Anatomical Structures Affected

Organ level (endocrine system, UBERON:0000949): - **Hypothalamus** (UBERON:0001898) — TIDA neurons; site of feedback failure. - **Anterior pituitary / adenohypophysis** (UBERON:0001885) — disinhibited lactotrophs; the source of excess prolactin, yet **structurally normal (no adenoma)**. - **Mammary gland / breast** (UBERON:0001911) — prolactin-resistant; agalactia. - **Ovary / gonads** (UBERON:0000992) and **uterus/decidua** (UBERON:0000995) — reproductive dysfunction, implantation failure (model).

Tissue/cell level: pituitary lactotrophs (CL:0002313); hypothalamic dopaminergic neurons (CL:0000700); mammary luminal epithelium (CL:0002326); granulosa/corpus luteum cells.

Subcellular level (GO Cellular Component): plasma membrane receptor (GO:0005886); downstream cytoplasmic JAK2 and nuclear STAT5 (GO:0005634 nucleus for STAT5 transcriptional action).

Localization/lateralization: Bilateral/systemic (endocrine); no lateralization.

---

8. Temporal Development

- **Onset:** Insidious, typically recognized in **adult reproductive age**, when menstrual disturbance, infertility, or postpartum lactation failure prompts evaluation.
- **Progression: Stable/chronic**, non-progressive. There is no mass lesion to grow and no evidence of neurodegeneration. Prolactin elevation is persistent.
- **Course:** Lifelong (germline). The heterozygous-mouse lactation phenotype is notably **age- and parity-dependent** — near-complete lactation failure after a first pregnancy when mated young (6–8 wk), attenuated when mated at 20 wk, and absent after a second pregnancy [PMID: 10935023] — suggesting biological windows (parity/age) can modulate expressivity.
- **Remission/critical periods:** No spontaneous remission of the genetic defect; reproductive windows are the main intervention periods.

---

9. Inheritance and Population

Epidemiology. No reliable prevalence/incidence figures exist for the *PRLR*-specific disease — it is **ultra-rare**, described in only a few kindreds since 2013. (Hyperprolactinemia overall is common, but the genetic *PRLR* subset is a tiny fraction.) Germline *PRLR* mutation is **not** a major cause of sporadic prolactinoma PMID: 26641246.

Inheritance pattern. Both documented: **autosomal dominant** (heterozygous His188Arg) and **autosomal recessive/biallelic** (compound-het Pro269Leu/Arg171*).

Penetrance / expressivity. Appears high for the biochemical trait (hyperprolactinemia) in reported dominant kindreds but with **variable clinical expressivity** (menstrual/fertility/lactation phenotypes vary). Model data imply **gene-dosage and age/parity-dependent** expressivity.

Anticipation, mosaicism, founder effects, consanguinity, carrier frequency: Not established / insufficient data. Variants appear private to families.

Demographics. Reported patients are predominantly **female** (ascertainment via reproductive phenotypes); male carriers may be underrecognized given subtler phenotypes. No defined ethnic/geographic clustering.

---

10. Diagnostics

Diagnostic strategy (diagnosis of exclusion → genetic confirmation):

1. **Confirm true hyperprolactinemia** — repeat serum prolactin; prolactin threshold >20 ng/mL (men) / >25 ng/mL (women) for non-physiological elevation [PMID: 39355173].
2. **Exclude macroprolactinemia** by PEG precipitation (big-big prolactin/anti-PRL IgG complexes). Macroprolactinemia accounts for **10–25%** of hyperprolactinemic patients and ~**3.7%** of the general population PMID: 23304187; in one Chinese cohort it was **22.9%** of hyperprolactinemic patients PMID: 34620143. Clinical features alone cannot reliably distinguish it [PMID: 22313536; 21527183].
3. **Exclude secondary causes** — drugs (dopamine antagonists), hypothyroidism, renal failure, pregnancy, chest-wall stimulation — per the Endocrine Society guideline PMID: 21296991.

4. **Pituitary MRI** — must be **NORMAL** (no micro-/macroadenoma). MRI is essential because prolactin level correlates imperfectly with adenoma presence (optimal cutoff ~38.7 ng/mL in one series) [PMID: 39792709].
5. **Genetic testing** — single-gene *PRLR* sequencing or **whole-exome sequencing (WES)** to identify the pathogenic variant [PMID: 24195502; 39355173]. WES has been the discovery modality in reported families.

Clinical lab/ontology: serum prolactin (LOINC 15081-0). Differential diagnosis includes prolactinoma, macroprolactinemia, drug-induced hyperprolactinemia, hypothyroidism, PCOS (where hyperprolactinemia may coexist or mask PCOS — PCOS is itself a diagnosis of exclusion) [PMID: 21388855].

Omics/biopsy/screening: No specific omics diagnostic; no biopsy; no population newborn screening. **Cascade genetic testing** of relatives is appropriate once a familial variant is identified.

---

11. Outcome / Prognosis

- **Survival/mortality: Not reduced** by the disease itself; it is non-malignant and non-life-threatening. No disease-specific mortality.
- **Morbidity/function:** Reproductive morbidity (infertility, menstrual disturbance, agalactia) and associated psychosocial burden.
- **Complications:** Chronic hyperprolactinemia generally (across causes) has been associated with sequelae such as hypogonadism and, in dopamine-agonist-treated cohorts, a modestly increased ischemic-stroke signal [PMID: 39924868] — but these data are not specific to the *PRLR* form and treatment context differs.
- **Recovery/prognostic factors:** Prognosis is favorable; the condition is stable. Because peripheral tissues are resistant, standard dopamine-agonist lowering of prolactin may not reverse end-organ symptoms the way it does in prolactinoma — a key prognostic/therapeutic distinction (inferred from resistance mechanism).

---

12. Treatment

There is no cure; management is supportive and reproductive-endocrine, not tumor-directed. Because the defect is receptor *resistance*, lowering prolactin pharmacologically does not restore prolactin signaling at resistant end-organs.

- **Dopamine agonists** (cabergoline — NCIT:C1868; bromocriptine — NCIT:C372) are first-line for hyperprolactinemia generally and may normalize prolactin levels [PMID: 21296991; 40966237], but their benefit in a *resistance* disorder is limited to lowering the circulating level rather than correcting end-organ non-response. Cabergoline is generally preferred for efficacy/tolerability; intolerance occurs in 3–12% [PMID: 37027090]. Dopamine agonists carry impulse-control-disorder risk (7.5–46% in prolactinoma/hyperprolactinemia cohorts) [PMID: 40316139].
- **Reproductive management:** assisted reproduction/fertility support for infertility; counseling regarding inability to breastfeed (agalactia).
- **Gene/cell/RNA/targeted/immuno-therapies:** None approved or in trials for this indication. No surgical role (no tumor).
- **Pharmacogenomics:** None specific.

Guideline anchor: Endocrine Society clinical practice guideline on hyperprolactinemia PMID: 21296991; Pituitary Society prolactinoma guidelines for differential context [PMID: 16886971].

13. Prevention

- **Primary prevention:** Not applicable (germline).
- **Secondary prevention: Cascade genetic screening** of at-risk relatives after a familial *PRLR* variant is identified; early recognition avoids misdiagnosis as prolactinoma and unnecessary imaging/pharmacotherapy.
- **Genetic counseling:** Essential — discuss autosomal dominant vs biallelic transmission, variable expressivity, and reproductive planning (NSGC/ACMG frameworks; preimplantation/prenatal options could be discussed though rarely indicated given benign course).
- **Immunization / public health / environmental:** Not applicable.

14. Other Species / Natural Disease

- **Taxonomy / orthologs:** *Prlr* is conserved in mouse (NCBI Gene 19116; *Mus musculus*, NCBI:txid10090) and rat (*Rattus norvegicus*, NCBI:txid10116). High evolutionary conservation of the prolactin–PRLR–JAK2–STAT5 axis.
- **Natural disease:** No well-characterized spontaneous *PRLR*-deficiency hyperprolactinemia syndrome is catalogued in companion animals/wildlife (OMIA search not definitive). Prolactin/PRLR variation in livestock affects reproduction/lactation traits but is not framed as this disease.
- **Comparative biology:** Engineered mouse knockouts (below) are the main cross-species evidence; mechanisms (lactation, implantation, feedback) are strongly conserved.
- **Zoonotic potential:** None (genetic disease).

15. Model Organisms

Mouse (*Mus musculus*) — the principal model.

Model	Key phenotype	Relevance to human disease	Evidence
Prlr-null (homozygous KO)	Female **sterility** from complete implantation failure; irregular cycles, reduced fertilization, defective preimplantation development, no pseudopregnancy; ~50% of males infertile/subfertile	Establishes PRLR as essential reproductive/ lactation regulator; models peripheral-resistance phenotype	PMID: 9009200
Prlr heterozygous (+/−)	Near-complete **lactation failure after first pregnancy** (young dams), rescued at 20 wk or second pregnancy; smaller, less-branched mammary glands	Models autosomal-dominant, **gene-dosage- and age-dependent** human phenotype	PMID: 10935023
Conditional neuronal *Prlr* deletion	Prolactin drives TIDA dopaminergic neuron signaling (short-loop feedback)	Mechanistic basis for central disinhibition → hyperprolactinemia	PMID: 30759390
Neutralizing anti-PRLR antibody (pharmacologic)	Reversible infertility, mammary involution, impaired lactation — phenocopies KO	Validates PRLR blockade = disease phenotype	PMID: 26284426

Phenotype recapitulation: Excellent for reproductive/lactation features and for the central-feedback mechanism. **Limitations:** Mouse models were engineered knockouts rather than the human point mutation; homozygous sterility prevents some multigenerational study; human dominant hyperprolactinemia biochemistry is better captured mechanistically than quantitatively. Resources: MGI, IMPC, IMSR.

Mechanistic Model / Interpretation

The unifying model is a **two-compartment consequence of one molecular lesion**. A loss-of-function *PRLR* allele simultaneously (1) disables the hypothalamic brake on prolactin secretion and (2) deafens peripheral tissues to the hormone. Compartment 1 explains the paradox at the heart of the disease — **high prolactin with no tumor** — because the elevation is driven by lost dopaminergic feedback rather than autonomous adenoma secretion. Compartment 2 explains

why the clinical signs (amenorrhea, infertility, agalactia) are those of prolactin *deficiency of action* despite prolactin *excess in blood*. This resolves what would otherwise look contradictory and is the single most important interpretive point for a clinician: in familial *PRLR* hyperprolactinemia, the number is high but the hormone "isn't working."

The **bidirectionality** of *PRLR* reinforces the model's logic. If loss of signaling disinhibits prolactin and deafens tissues, then **gain** of signaling (constitutive receptors: I146L, I76V) should produce ligand-independent proliferation in prolactin-responsive epithelium — and indeed these variants cause benign breast fibroadenomas [PMID: 18779591; 19897676]. The same receptor, read in opposite directions, yields opposite diseases.

```
   LOSS of function            GAIN of function
(His188Arg, Arg171*)          (Ile146Leu, Ile76Val)
         │                             │
prolactin resistance          constitutive STAT5 activation
         │                             │
▼ feedback / ▼ action          ▲ proliferation
         │                             │
FAMILIAL HYPERPROLACTINEMIA    BENIGN BREAST FIBROADENOMAS
  (tumor-negative)
```

Evidence Base

PMID	Title (abbrev.)	Role
24195502	*Mutant prolactin receptor and familial hyperprolactinemia*	**Founding evidence**: His188Arg LOF disrupts ligand binding, abolishes JAK2-STAT5; segregates in 3 sisters; no tumor/MEN
36445946	*PRLR variants with diverse effects on signalling*	Confirms His188Arg inactivating; biallelic Pro269Leu/Arg171* with agalactia; notes activating Asn492Ile/prolactinoma
30759390	*Elevated prolactin drives TIDA neuron switch*	Mechanism: PRL short-loop feedback on dopaminergic neurons
9009200	*Null mutation of Prlr — reproductive defects*	KO mouse: sterility, implantation failure
10935023	*Mammary gland development in Prlr KO*	Dominant, dosage/age-dependent lactation failure in het mice
18779591	*Gain-of-function PRLR mutation in benign breast tumors*	Opposite pole: constitutive activity, fibroadenomas
19897676	*Two constitutively active PRLR variants (95 women)*	Confirms I146L and I76V constitutive activity
26641246	*Germline PRLR not major cause of sporadic prolactinoma*	Boundary: common PRLR variants functionally neutral
21296991	*Endocrine Society hyperprolactinemia guideline*	Diagnostic/therapeutic framework
23304187	*Macroprolactinemia significance*	Quantifies key differential (10–25%)
39355173	*Ser434Phe PRLR*	Additional candidate variant; WES approach

Key verified quotes: - "This substitution disrupted the high-affinity ligand-binding interface of the prolactin receptor, resulting in a loss of downstream signaling by Janus kinase 2 (JAK2) and signal transducer and activator of transcription 5 (STAT5)" — [PMID: 24195502]. - "an inactivating germline heterozygous PRLR variant (His188Arg) was reported in a kindred with hyperprolactinaemia, while an inactivating compound heterozygous PRLR variant (Pro269Leu/Arg171Stop) was identified in an individual with hyperprolactinaemia and agalactia" — [PMID:

36445946]. - "Homozygous females were sterile owing to a complete failure of embryonic implantation... Half of the homozygous males were infertile or showed reduced fertility." — [PMID: 9009200]. - "Heterozygous females showed almost complete failure of lactation following their first pregnancy when mated at 6-8 weeks of age." — [PMID: 10935023]. - "These neurons normally release dopamine to inhibit prolactin secretion and are activated by prolactin in a short-loop feedback manner." — [PMID: 30759390]. - "This sole substitution was sufficient to confer constitutive activity to the receptor variant" — [PMID: 18779591]. - "The prevalence of macroprolactinemia is 10-25% in patients with hyperprolactinemia and 3.7% in general population." — [PMID: 23304187].

Limitations and Knowledge Gaps

1. **Ultra-rarity:** Only a handful of kindreds/variants are published. Prevalence, incidence, penetrance estimates, sex ratio, and geographic/founder data are essentially unknown.
2. **Human central mechanism is inferred:** The TIDA-feedback disinhibition step is demonstrated in mouse; direct human hypothalamic evidence is lacking. Labeled as inference throughout.
3. **Variant interpretation:** Several *PRLR* missense variants are common and functionally neutral [PMID: 26641246]; candidate variants (e.g., Ser434Phe) need functional confirmation. Not all reported variants are ACMG-classified pathogenic.
4. **Therapeutics:** No disease-specific trials. Whether dopamine agonists meaningfully help a resistance disorder is untested; end-organ resistance predicts limited symptomatic benefit.
5. **Male and multigenerational phenotype** underdescribed due to ascertainment bias toward reproductive-age women.
6. **No omics/epigenetic/structural data** specific to the human disease.

Proposed Follow-up Experiments / Actions

1. **Functional assays** (JAK2/STAT5 reporter, ligand-binding) for all newly reported *PRLR* variants (e.g., Ser434Phe) to resolve pathogenicity per ACMG PS3.
2. **International kindred registry / GeneMatcher outreach** to aggregate cases, estimate penetrance/expressivity, and define male phenotype.

3. **Knock-in mouse (His188Arg)** modeling the exact human dominant allele to test whether it reproduces hyperprolactinemia (not just reproductive phenotypes) and to directly measure TIDA dopaminergic tone.
4. **Human hypothalamic-pituitary feedback testing** (dynamic prolactin/dopamine axis studies) in affected carriers to confirm the central-disinhibition mechanism.
5. **Therapeutic evaluation**: prospective assessment of whether prolactin-lowering alters reproductive outcomes in confirmed carriers, versus assisted-reproduction-first strategies.
6. **Structural modeling (AlphaFold/cryo-EM)** of His188Arg/Pro269Leu to map how each disrupts the ligand interface and dimerization.

Report compiled from 5 discovery iterations, 5 confirmed findings, and 39 reviewed papers. Evidence types: human clinical/genetic (kindreds, guidelines), model organism (mouse KO/het, antibody), in vitro functional (signaling assays), and computational/inferential (central-feedback extrapolation).
